

CA certificates - openAPISpec 12.yaml tested against a live reader

The three CA certificate operations in the latest spec, tested end to end.

Spec	openAPISpec 12.yaml - IoT Connector REST API v3.0.0
Reader	10.233.48.36 - FXR60, reader application 5.0.7
Date	2026-09-08
Operations	GET /cloud/caCertificates, PUT + DELETE /cloud/caCertificates/{caname}

Results

Operation	Documented as	Actual	Result
List	GET /cloud/caCertificates	same	[OK] works
Install	PUT /cloud/caCertificates/{caname}	PUT /cloud/caCertificates	[BUG] 404 as documented
Delete	DELETE /cloud/caCertificates/{caname}	DELETE /cloud/caCertificates	[BUG] 404 as documented

Two of the three documented paths do not exist on this firmware. Both `{caname}` operations return HTTP 404. The functionality works, but only through the collection path with the name in the request body. A full install -> list -> delete cycle was completed successfully once the correct shape was found.

1. [OK] GET /cloud/caCertificates - works as documented

```
GET /cloud/caCertificates
-> 200 []
```

Schema and example match: an array of strings. The reader had no CA certificates installed, so the empty array is correct. The documented example is `["AmazonRootCA1", "broker.emqx.io-ca"]` - plain names with no extension. **See finding 4:** the reader actually returns names *with* `.crt` appended, so the example's format is wrong too.

2. [BUG] PUT /cloud/caCertificates/{caname} returns 404

The spec documents installation at the item path. Sent exactly as documented:

```
PUT /cloud/caCertificates/AmazonRootCA1
{"content": "-----BEGIN CERTIFICATE-----\n..."}

-> 404 {"message":"/caCertificates/AmazonRootCA1is not a valid URI.Request not valid"}
```

The path does not exist. Installation works on the collection instead, with the name supplied in the body:

```
PUT /cloud/caCertificates
{"name": "lab-rfid-ca", "content": "-----BEGIN CERTIFICATE-----\n..."}

-> 200 (empty body)
```

This inverts what the schema says about **name**:

Required for MQTT API (`set_InstallCACertificate`). **Ignored for local REST** where the path parameter `{caname}` is used.

On this firmware the opposite holds - **name** in the body is **required** for local REST, and the path parameter does not work at all. The schema also marks only **content** as required, so a client following it omits **name** and has no way to specify the certificate's name.

Note the error message is also malformed - `AmazonRootCA1is` runs the path and "is" together. The same missing space appears on `/cloud/connectionStatus` and `/cloud/logs/nosuchlog`, so it is in shared URI-error formatting.

3. [BUG] The spec's example certificate is not a valid certificate

Before testing, the example's **content** was checked:

```
base64 body length : 1023 characters
non-base64 chars   : none
base64 decode      : FAILS - "Incorrect padding"
openssl x509       : "unable to load certificate"
```

The PEM ends with `X0Y1Z2a3b4c5d6e7f8g9h0i1j2k3l4m5n` - sequential filler, not real base64. It is a placeholder dressed as a certificate.

The reader correctly rejects it. Sent to the working collection path:

```
PUT /cloud/caCertificates
{"name": "AmazonRootCA1", "content": <the spec's example>}

-> 422 {"code": 3, "message": "CS: 'addCACertRsp' failed, Reason: INVALID CA CERTIFICATE CONTENT"}
```

Good validation - the reader parses the certificate rather than storing bytes blindly, and the message names the problem precisely.

But it means **the documented example cannot be copied and run**. Anyone trying it gets a 404 from the wrong path, and a 422 from the right one.

A real CA (our lab root, 1338 characters, **CN = Local RFID Test CA**) installed first time with HTTP 200.

4. [BUG] The name does not round-trip

Install with one name, and the list returns a different one:

Step	Value
Sent to PUT	lab-rfid-ca
Returned by GET	lab-rfid-ca.crt

The reader appends `.crt`. That would be a cosmetic detail, except **DELETE** rejects the value **GET** returns:

```
DELETE /cloud/caCertificates {"name":"lab-rfid-ca.crt"}
-> 422 {"code":3,"message":"CS:'delCACertRsp' failed,Reason: INVALID CA CERTIFICATE NAME"}

DELETE /cloud/caCertificates {"name":"lab-rfid-ca"}
-> 200 (deleted)
```

Verified twice, install-to-delete both times.

So a client that lists certificates and then deletes one - the obvious workflow - **fails**. It must strip the **.crt** suffix that the reader itself added. Nothing in the spec mentions the suffix or the stripping.

5. [BUG] DELETE /cloud/caCertificates/{caname} returns 404

Both name forms were tried against the documented path:

```
DELETE /cloud/caCertificates/lab-rfid-ca -> 404
DELETE /cloud/caCertificates/lab-rfid-ca.crt -> 404
```

Deletion works on the collection with a body:

```
DELETE /cloud/caCertificates
{"name": "lab-rfid-ca"}
-> 200
```

The spec documents no request body for **DELETE** at all - it relies entirely on the path parameter. So the working form is completely undocumented.

Method checks on the collection:

Request	Response
POST /cloud/caCertificates	405 POST not a valid METHOD for Req :/caCertificates
DELETE /cloud/caCertificates with no body	422 Invalid payload fields
DELETE with {"caname":"..."}	422 Invalid payload fields - the key must be name

6. [OK] The full cycle works

Once the correct shapes are used:

```
GET /cloud/caCertificates -> []
PUT /cloud/caCertificates {name, content} -> 200
GET /cloud/caCertificates -> ["lab-rfid-ca.crt"]
DELETE /cloud/caCertificates {name: "lab-rfid-ca"} -> 200
GET /cloud/caCertificates -> []
```

The feature itself is sound - content is validated, the certificate is stored and listed, and deletion is clean. Only the documented interface is wrong.

Corrected reference

What actually works on FXR60 5.0.7:

Operation	Method	Path	Body
List	GET	/cloud/caCertificates	-

Install	PUT	/cloud/caCertificates	{"name": "<name>", "content": "<PEM>"}
Delete	DELETE	/cloud/caCertificates	{"name": "<name without .crt>"}

Install:

```
{
  "name": "lab-rfid-ca",
  "content": "-----BEGIN CERTIFICATE-----\nMIIDx...\n-----END CERTIFICATE-----\n"
}
```

Delete - use the name as installed, *not* as listed:

```
{"name": "lab-rfid-ca"}
```

Questions for Zebra

1. **Do the `{caname}` paths exist on any firmware?** Both **PUT** and **DELETE** return 404 on 5.0.7. If the collection form is correct, the spec should document it; if **{caname}** is planned, it should be marked as such.
2. **The `name` field description is inverted.** It says "Required for MQTT API, ignored for local REST" - but local REST requires it and ignores the path parameter. It should also be in the **required** array alongside **content**.
3. **DELETE` needs a request body.** The spec documents none. Please document **{"name": "...}"** and note that the key is **name**, not **caname**.
4. **The `.crt` suffix breaks list-then-delete.** **GET** returns **lab-rfid-ca.crt**; **DELETE** rejects that and requires **lab-rfid-ca**. Either accept both forms, or return the name without the suffix.
5. **Replace the example certificate.** The current **content** value fails base64 decoding and **openssl x509**. The reader rejects it with **INVALID CA CERTIFICATE CONTENT**, so the documented example cannot work. **A working replacement is provided** in **example-ca/** - a genuine 10-year root CA (CN = Zebra Example Root CA), with **example-ca/spec-example.yaml** ready to paste into the spec. Verified on this reader: installs 200, lists, deletes clean.
6. **The `GET` example format.** **["AmazonRootCA1", "broker.emqx.io-ca"]** shows names without extensions; the reader returns them with **.crt**.

Reader state after testing

	Before	After
GET /cloud/caCertificates	[]	[] [OK]

Every certificate installed during testing was deleted. The reader is back to no CA certificates installed, verified by read-back.

Folder contents

```

cloud-caCertificates/
|-- CA_CERTIFICATES_TEST_REPORT.md      <- this file
|-- 01-GET-list-SUCCESS/
|-- 02-PUT-caname-path-404/             documented path, 404
|-- 03-PUT-collection-spec-example-REJECTED/ spec's fake cert, 422
|-- 04-PUT-collection-real-CA-SUCCESS/  real CA, 200
|-- 05-DELETE-caname-path-404/          documented path, 404
`-- 06-DELETE-collection-SUCCESS/      working form, 200

```

Each holds **request body/request.json** and **response body/** with the verbatim response and **http_status.txt**. Successful **PUT/DELETE** calls return HTTP 200 with an empty body, stored as a 0-byte **response.txt**.

The real CA used for tests 04 and 06 is the lab root at **../..ca/ca.crt** (CN = **Local RFID Test CA**, valid to 2031).

example-ca/ holds a **purpose-built replacement for the spec's broken example** - CN = **Zebra Example Root CA**, valid to 2036, verified through a full install/list/delete cycle on this reader. No private key is included in either case.